TEST REPORT

No. : GZMR240400842204

Date : May 11, 2024

Scan to see the report

GZMR240400842204

Verification: check.sgsonline.com.cn

CUSTOMER NAME: NINGBO ZERO ENGINEERING PLASTICS CO.,LTD

ADDRESS: #5.#6.NO.406,JIANGHU ROAD,JIANGBEI DISTRICT,NINGBO

Sample Name : PEEK

Above information and sample(s) was/were submitted and confirmed by the client. SGS, however, assumes no responsibility to verify the accuracy, adequacy and completeness of the sample information provided by client.

****

SGS Ref. No.	:	NBIN2404000821PL04
Date of Receipt	:	Apr 01, 2024
Testing Start Date	:	Apr 01, 2024
Testing End Date	:	May 11, 2024
Test result(s)	:	For further details, please refer to the following page(s) (Unless otherwise stated the results shown in this test report refer only to the sample(s) tested)

Signed for
SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch

May Huo
Authorized signatory

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TEST REPORT

Summary of Results:

No.	Test Item	Test Method	Result		Conclusion
1	Vertical Burning	UL 94-2023 Section 8	V-0		/
2	Surface Resistance	IEC 62631-3-2:2023	6.97E+13Ω		/
3	Volume Resistance	IEC 62631-3-1:2023	2.64E+14Ω		/
4	Tensile Test	ISO 527-2:2012	Tensile Strength	96.3MPa	/
			Tensile Strain at Break	4.4%	
5	Flexural Test	ISO 178:2019 Method A	Flexural Strength	170MPa	/
			Flexural Modulus	4010MPa	
6	Rockwell Hardness	ISO 2039-2:1987	113 L		/
7	Specific Gravity	ISO 1183-1:2019 Method A	1.301		/
8	Water Absorption	ISO 62:2008 Method 1	0.04%		/
9	Mean Coefficient of Linear Thermal Expansion	ISO 11359-2:2021 Method A	0℃~140℃	53×10^{-6} K^{-1}	/
			160℃ ~200℃	124×10^{-6} K^{-1}	
10	Deflection Temperature under Load	ISO 75-2:2013 Method A	160℃		/
11	Melting Temperature (Tm)	ISO 11357-3:2018	See result		/
12	Relative Permittivity	IEC 62631-2-1:2018	3.204		/

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13	Breakdown Voltage and Dielectric Strength	IEC 60243-1:2013	See result	/
14	Charpy Impact Strength	ISO 179-1:2023	6.88kJ/m^2 C (complete break)	/

Note: Pass : Meet the requirements;

Fail : Does not meet the requirements;

/ : Not Apply to the judgment.

Original Sample Photo:

Original Sample-Sample D

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TEST REPORT

1.Test Item: Vertical Burning

Sample Description: Sheet

Test Method: UL 94-2023 Section 8

Test Condition:

Specimen: 125mm×13.0mm×3.7mm

Test Result:

Conditioning	As received :(23±2)℃, (50±5)%RH, 48h					Ageing: (70±2)℃, 168h→ Desiccator, (23±2)℃, 4h				
Specimen	1	2	3	4	5	1	2	3	4	5
Afterflame time after the first flame application (s), t_1	0	0	0	0	0	0	0	0	0	0
Afterflame time after the second flame application (s), t_2	0	0	0	0	0	0	0	0	0	0
Afterglow time after the second application (s), t_3	0	0	0	0	0	0	0	0	0	0
Total afterflame time for any condition set (s), t_1+t_2	0					0				
Afterflame time plus afterglow time after the second flame application (s), t_2+t_3	0	0	0	0	0	0	0	0	0	0
Whether or not specimen burn up to the holding clamp	No	No	No	No	No	No	No	No	No	No
Whether or not the cotton was ignited by flaming particles or drops	No	No	No	No	No	No	No	No	No	No

Classification: V-0 (See table 1)

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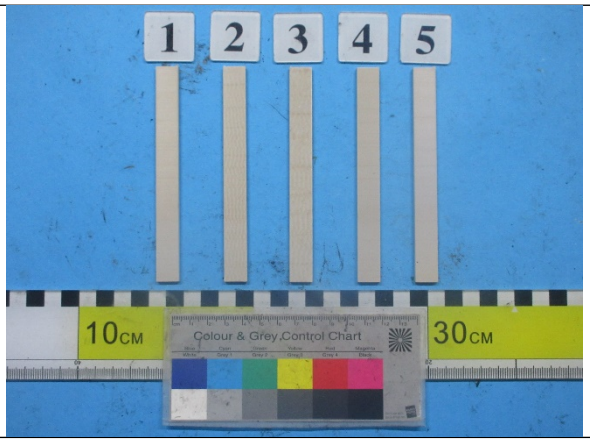
Table 1

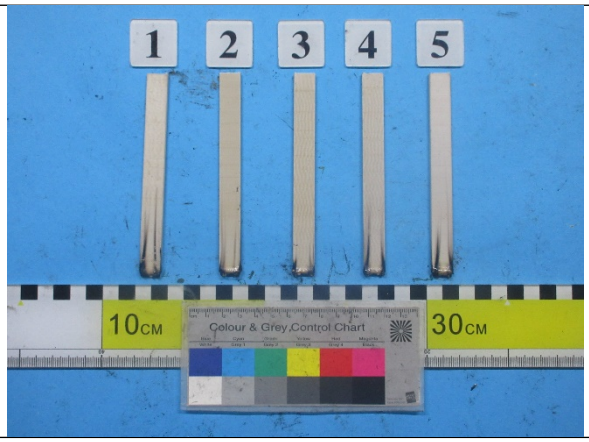
Criteria conditions	V-0	V-1	V-2
Afterflame time for each individual specimen t_1 and t_2	≤10s	≤30s	≤30s
Total afterflame time for any condition set (t_1 plus t_2 for the 5 specimens)	≤50s	≤250s	≤250s
Afterflame plus afterglow time for each individual specimen after the second flame application (t_2+t_3)	≤30s	≤60s	≤60s
Afterflame or afterglow of any specimen up to the holding clamp	No	No	No
Cotton indicator ignited by flaming particles or drops	No	No	Yes

Note: Test specimens were cut from the sample.

Test Photo:



Before Test-Sample D-As received



After Test-Sample D-As received

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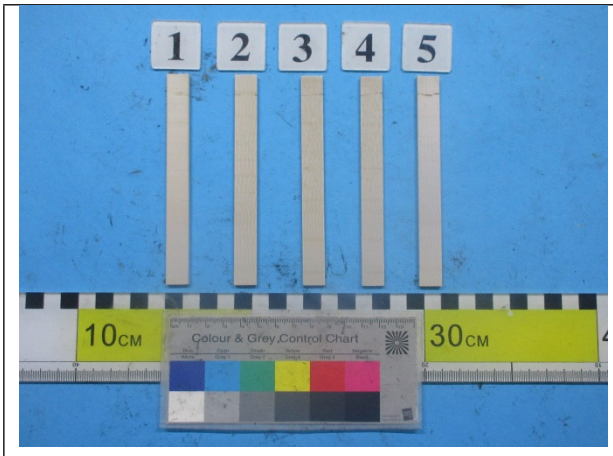	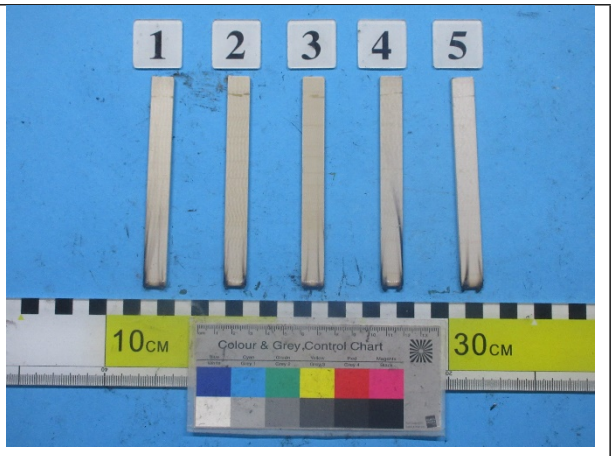
Before Test-Sample D-After ageing	After Test-Sample D-After ageing

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Horizontal/Vertical Flame Chamber	HVUL2	GZMR-PL-E297	2023-04-20	2024-04-19
Oven	UF750	GZMR-AG-E181	2023-08-24	2024-08-23

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2.Test Item: Surface Resistance

Sample Description: Sheet

Test Method: IEC 62631-3-2:2023

Test Condition:

Precondition: (23±2) ℃, (50±5) %RH, 96 h

Test electrode: D_1=50 mm, D_2=70 mm

Test voltage: 100 Vdc

Test duration: 1 min

Lab environmental: (23±2) ℃, (50±5) %RH

Test Result:

Specimen No.	Surface Resistance R_s (Ω)
1	4.56E+13
2	7.17E+13
3	2.10E+14
4	6.97E+13
5	4.93E+13
Median	6.97E+13

Note:

1) Test specimens were cut from the sample.
2) E+n refers to 10^n, for example, 1.00E+03 is equivalent to 1.00×10^3.

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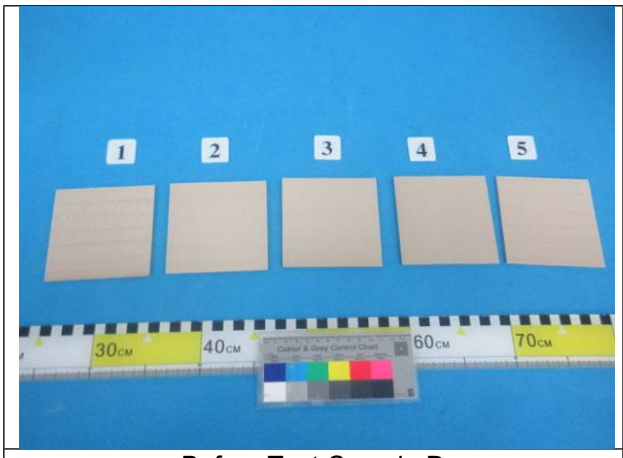	
Before Test-Sample D	After Test-Sample D

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
High Resistance Meter	4339B	GZMR-PL-E150	2023-05-16	2024-05-15

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TEST REPORT

3.Test Item: Volume Resistance

Sample Description: Sheet

Test Method: IEC 62631-3-1:2023

Test Condition:

Precondition: (23±2) ℃, (50±5) %RH, 96 h

Test electrode: D_1=50 mm, D_2=70 mm

Test voltage: 100 Vdc

Test duration: 1 min

Lab Environmental: (23±2) ℃, (50±5) %RH

Test Result:

Specimen No.	Thickness (mm)	Volume Resistance R_v (Ω)
1	3.69	2.07E+14
2	3.61	3.57E+14
3	3.74	2.64E+14
4	3.67	3.26E+14
5	3.64	8.79E+13
Median	/	2.64E+14

Note:

1) Test specimens were cut from the sample.
2) E+n refers to 10^n, for example, 1.00E+03 is equivalent to 1.00×10^3.

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	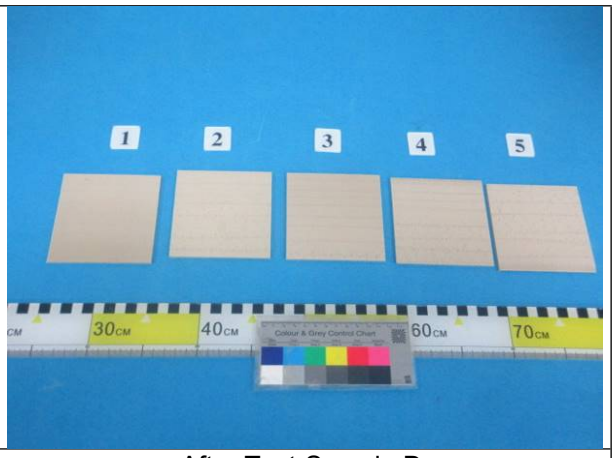
Before Test-Sample D	After Test-Sample D

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
High Resistance Meter	4339B	GZMR-PL-E150	2023-05-16	2024-05-15
Thickness meter	S112MXB	GZMR-RL-E115-01	2023-10-31	2024-10-30

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TEST REPORT

4.Test Item: Tensile Test

Sample Description: Sheet

Test Method: ISO 527-2:2012

Test Condition:

Specimen type: 1B

Specimen thickness: 3.67mm

Testing speed: 5mm/min

Gauge length: 50mm

Initial distance between grips: 115mm

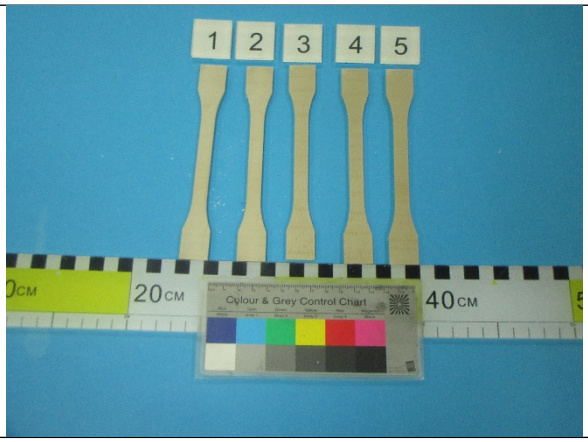
Lab Environmental Condition: (23±2)℃, (50±5)%RH

Test Result:

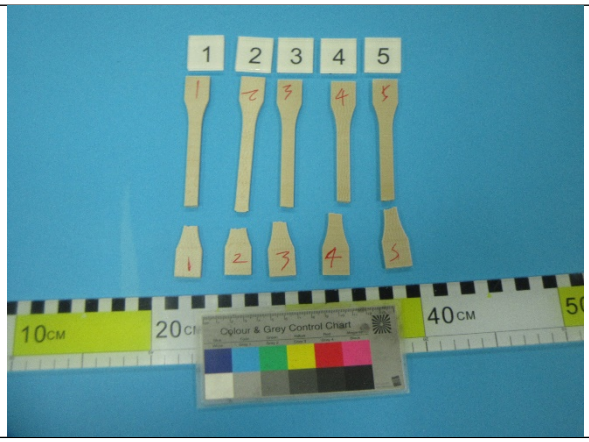
Test Item	Test Result
Tensile Strength	96.3MPa
Tensile Strain at Break	4.4%

Note: Test specimens were cut from the sample.

Test Photo:



Before Test-Sample D



After Test-Sample D

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TEST REPORT

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Automation Universal Testing Machine	Z020	GZMR-PL-E399	2023-05-15	2024-05-14

5.Test Item: Flexural Test

Sample Description: Sheet

Test Method: ISO 178:2019 Method A

Test Condition:

Specimen: 80mm×10.12mm×3.72mm

Testing speed: 2mm/min

Span: 59mm

Lab Environmental Condition: (23±2)℃, (50±5)%RH

Test Result:

Test Item	Test Result
Flexural Strength	170MPa
Flexural Modulus	4010MPa

Note: Test specimens were cut from the sample

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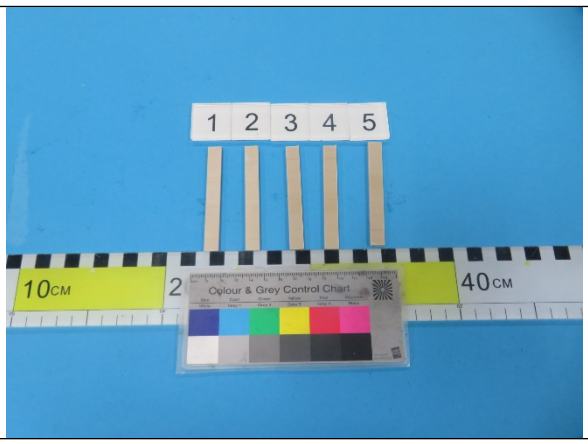	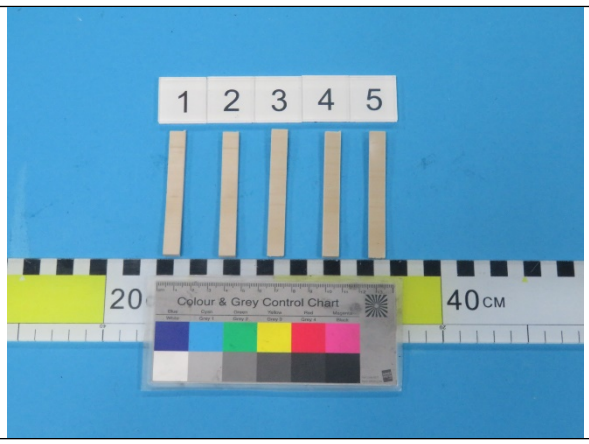
Before Test-Sample D	After Test-Sample D

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Universal Testing Machine	Z1.0	GZMR-PL-E252	2023-11-01	2024-10-31

6.Test Item: Rockwell Hardness

Sample Description: Sheet

Test Method: ISO 2039-2:1987

Test Condition:

Specimen thickness: 7.17mm (2 layers plied up)

Lab Environmental Condition: 23 ± 2 ℃, 50 ± 5 %RH

Test Result:

Test Item	Test Result
Rockwell Hardness	113 L

Note: Test specimens were cut from the sample.

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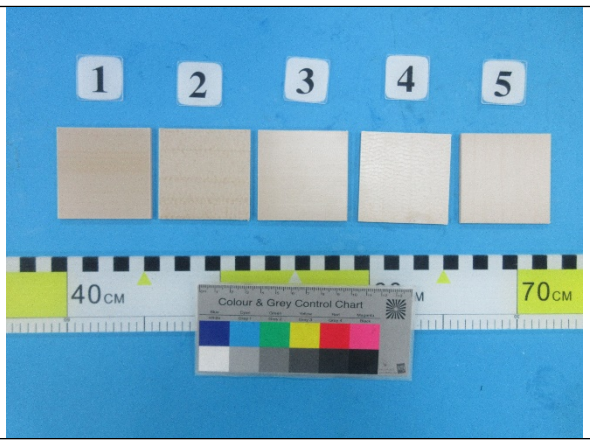	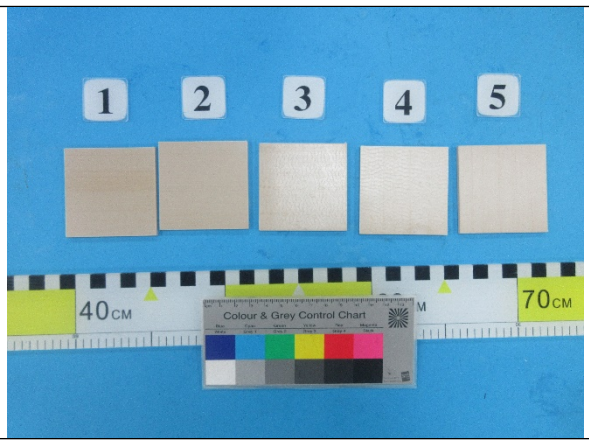
Before Test-Sample D	After Test-Sample D

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Rockwell Hardness Testing Machine	HR-521	GZMR-MT-E080	2024-01-04	2025-01-03

7.Test Item: Specific Gravity

Sample Description: Sheet

Test Method: ISO 1183-1:2019 Method A

Test Condition:

Absolute alcohol, (23±0.5)℃

Lab Environmental Condition: (23±2)℃, (50±5)%RH

Test Result:

Test Item	Test Result
Specific Gravity	1.301

Note: Test specimens were cut from the sample

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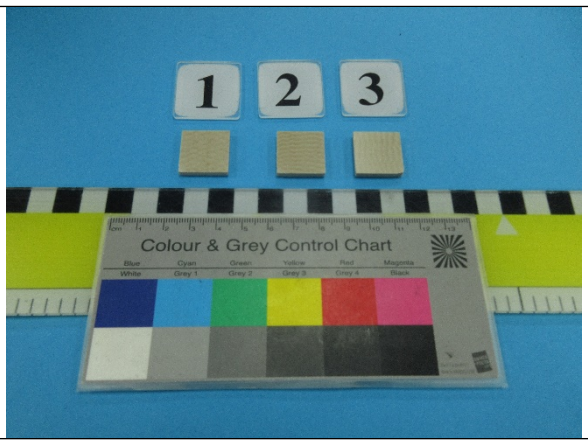	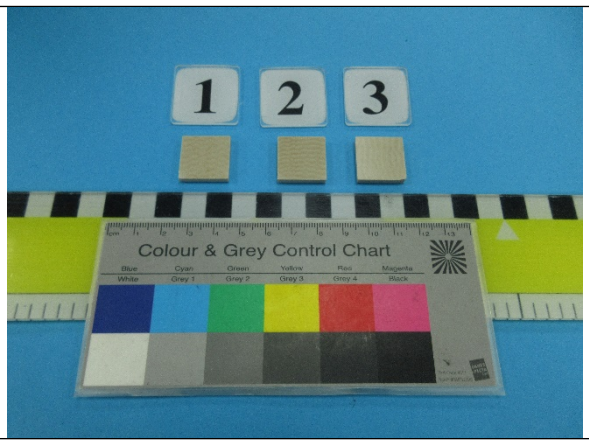
Before Test-Sample D	After Test-Sample D

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Electron Balance	XPE204	GZMR-PL-E004-06	2023-05-15	2024-05-14

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8.Test Item: Water Absorption

Sample Description: Sheet

Test Method: ISO 62:2008 Method 1

Test Condition:

Specimen: 62mm×62mm×3.8mm

Drying condition: 50±2℃, 24h

Immersion condition: 23±2℃, 24h

Lab Environmental Condition: 23±2℃, 50±5%RH

Test Result:

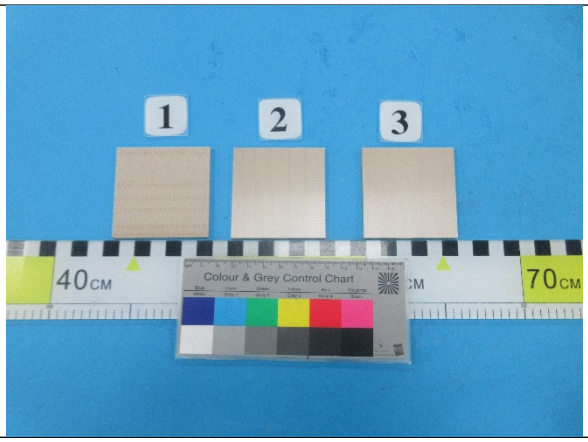
Test Item	Test Result
Water Absorption	0.04%

Note:

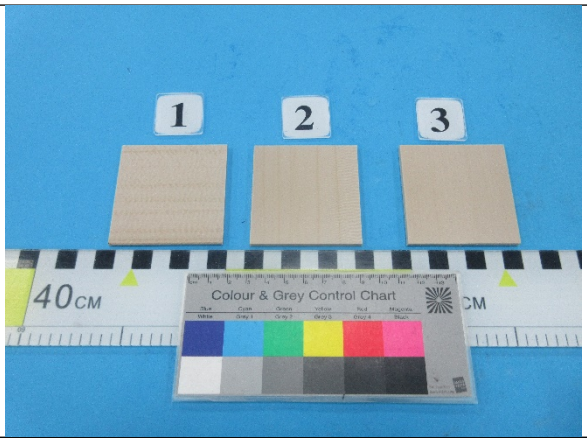
1.Water absorption, %=(Mass after immersion-Mass after drying and before immersion)/Mass after drying and before immersion×100

2.Test specimens were cut from the sample.

Test Photo:



Before Test-Sample D



After Test-Sample D

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Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Electron Balance	XPE404S	GZMR-PL-E004-07	2023-08-11	2024-08-10
Natural Ventilation Oven	UN110	GZMR-PL-E265	2023-05-12	2024-05-11

9.Test Item: Mean Coefficient of Linear Thermal Expansion

Sample Description: Sheet

Test Method: ISO 11359-2:2021 Method A

Test Condition:

Specimen: 9.97mm×5.43mm×3.42mm

Rate of temperature: 5 ℃/min

Load: 50mN

Flow rate(N_2): 50 ml/min

Test temperature: 0 ℃~200 ℃

Test mode: Compression

Test direction: Length

Number of specimens tested: 1

Lab Environmental Condition: (23 ± 2) ℃, (50 ± 5) % RH

Test Result:

Test Item	Test Result	
Mean Coefficient of Linear Thermal Expansion	0℃~140℃	53×10^{-6} K^{-1}
	160℃~200℃	124×10^{-6} K^{-1}

Note:

1. Test specimens were cut from the sample.
2. Test result was obtained after annealing.

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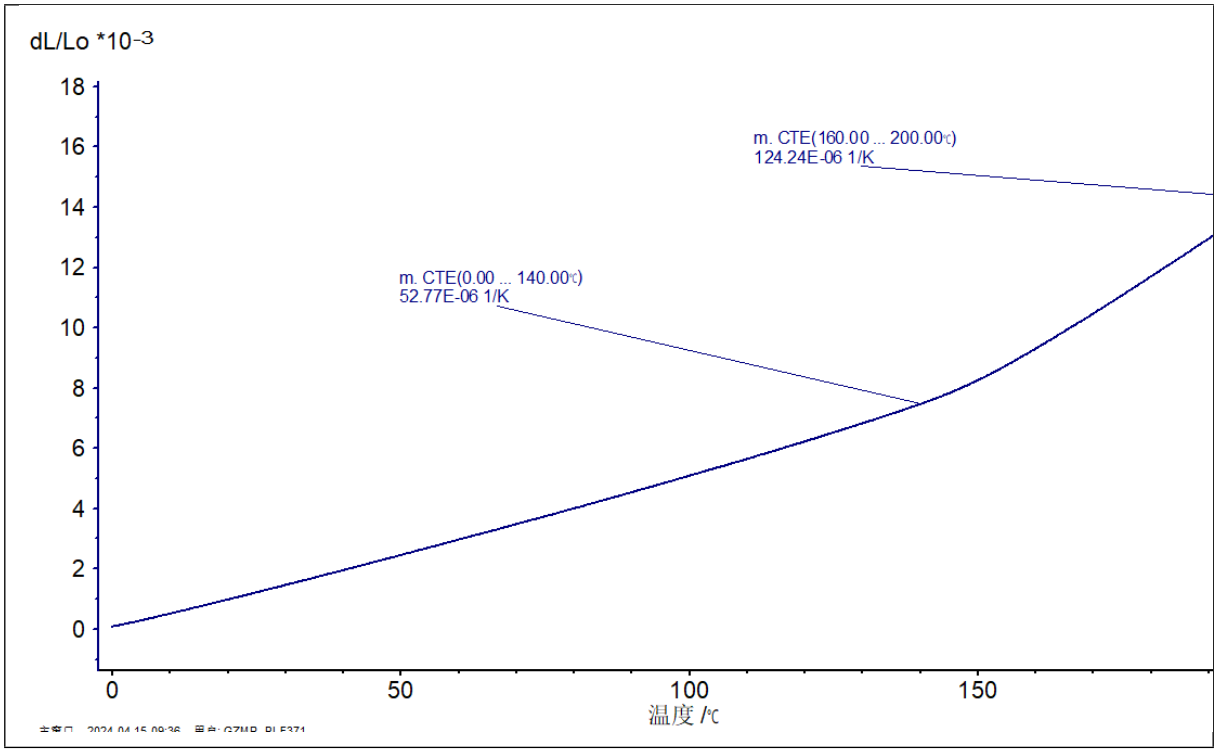
Curve – Length direction



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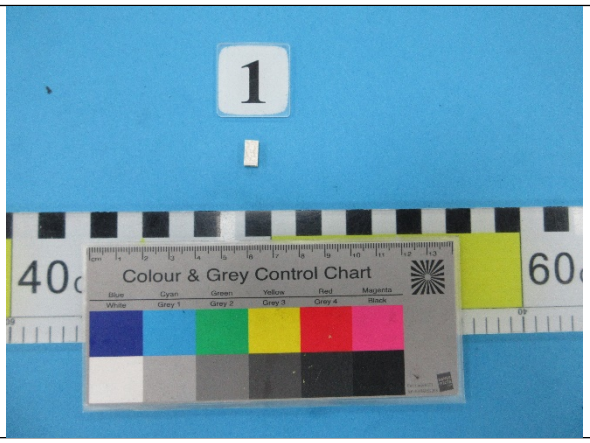	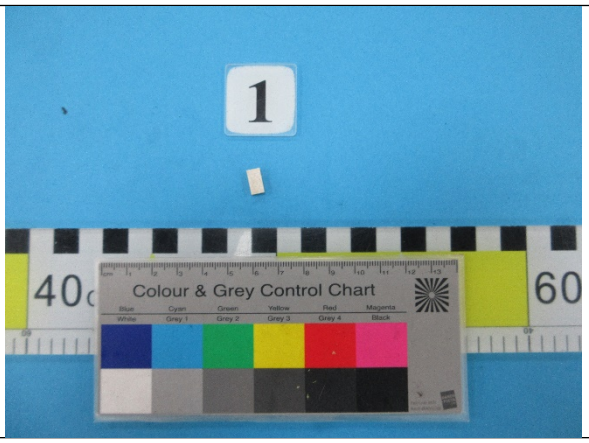
Before Test-Sample D	After Test-Sample D

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Thermo Mechanical Analyzer	TMA402F3	GZMR-PL-E371	2023-06-06	2024-06-05

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10. Test Item: Deflection Temperature under Load

Sample Description: Sheet

Test Method: ISO 75-2:2013 Method A

Test Condition:

Specimen: 80mm×10.22mm×3.74mm

Heat-transfer media: Silicone oil

T_f1.8

Rate of temperature:120℃/h

Span: 64mm

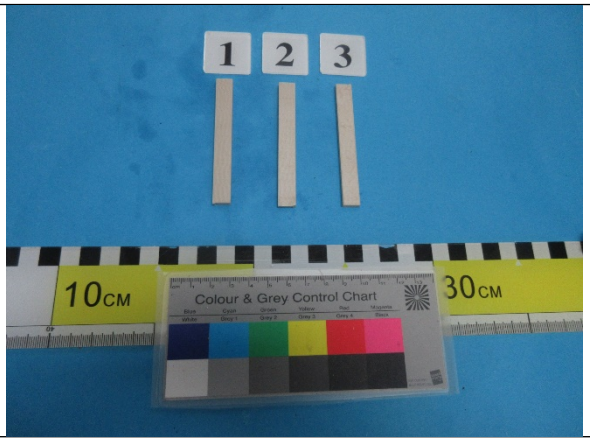
Lab Environmental Condition: (23±2)℃，(50±5)%RH

Test Result:

Test Item	Test Result
Deflection Temperature under Load	160℃

Note: Test specimens were cut from the sample.

Test Photo:



Before Test-Sample D

After Test-Sample D

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Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
HDT and VICAT Tester	IC 6	GZMR-PL-E348	2023-06-21	2024-06-20

11.Test Item: Melting Temperature (T_m)

Sample Description: Sheet

Test Method: ISO 11357-3:2018, analysis was performed by DSC.

Test Condition:

Type of sample container: Al;

Purge gas flow rate (Nitrogen: 99.999%): 50ml/min;

Temperature range: 40~380℃;

Ramp: 10℃/min.

Test Result:

Test Item	First heating scan (DSC curve see Fig.1)	Second heating scan (DSC curve see Fig.2)
Melting extrapolated onset temperature, T_{eim}	320.6℃	327.4℃
Melting peak temperature, T_{pm}	340.8℃	341.1℃
Melting extrapolated end temperature, T_{efm}	348.4℃	347.6℃

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Fig.1

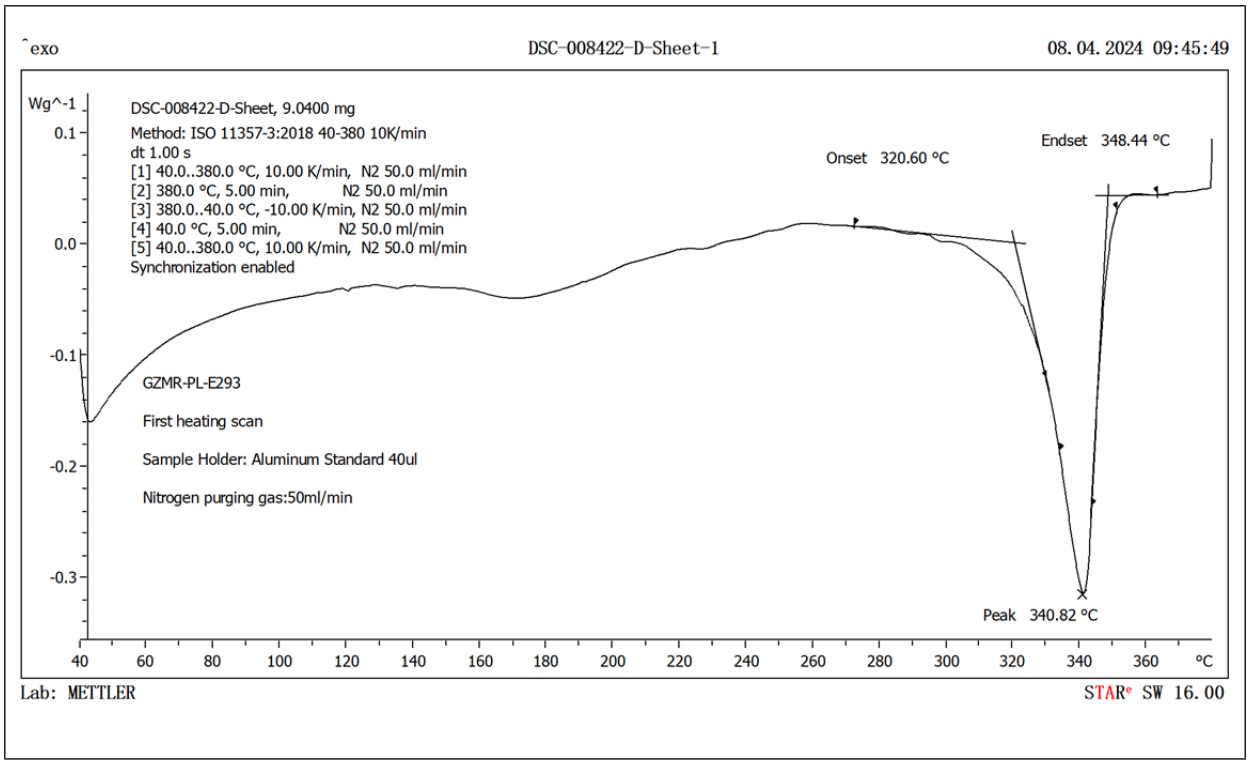

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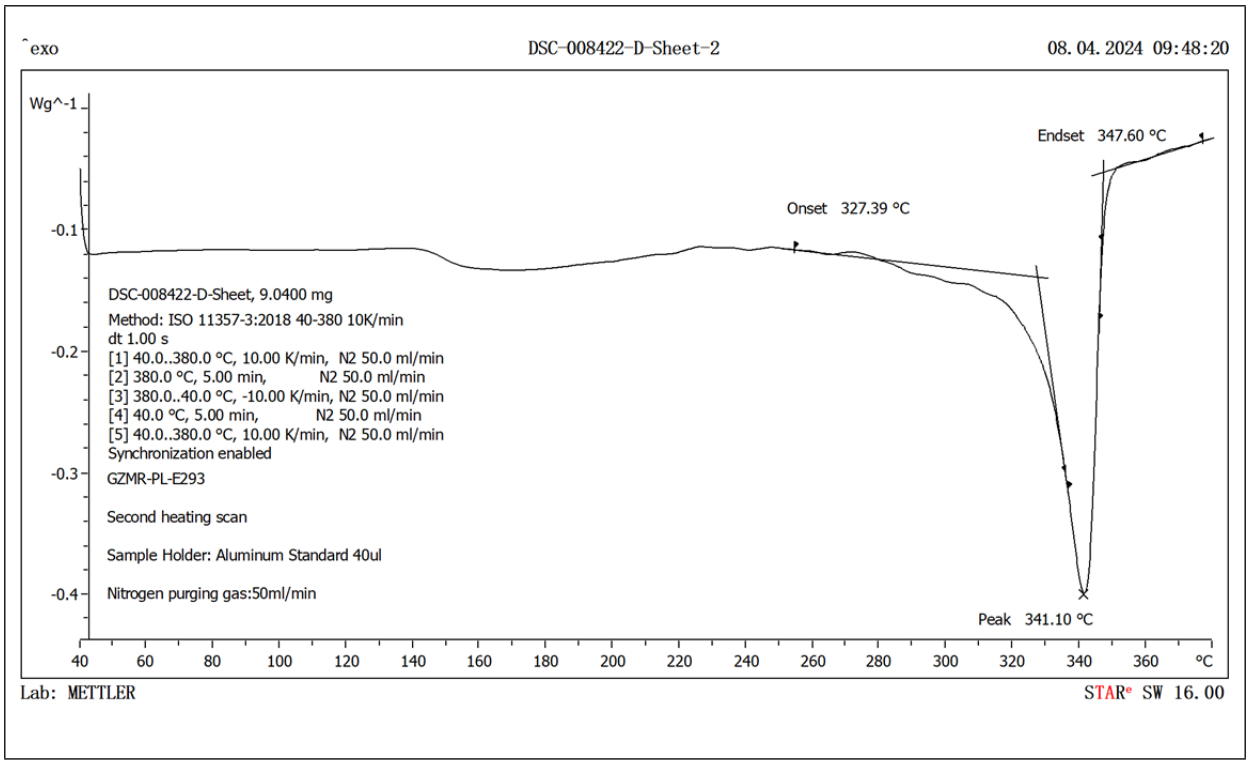
Fig.2



Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
DSC	DSC 3	GZMR-PL-E293	2024-03-05	2026-03-04

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TEST REPORT

12.Test Item: Relative Permittivity

Sample Description: Sheet

Test Method: IEC 62631-2-1:2018

Test Condition:

Precondition: (23±2) ℃, (50±5) %RH, 96 h

Test Electrode: Parallel electrode Φ38 mm

Test Frequency: 1 M Hz

Lab Environmental: 22.3 ℃, 51.6 %RH

Test Result:

Specimen No.	Thickness (mm)	Capacitance C_p (pF)	Relative Permittivity ε_r	Dissipation Factor D
1	3.50	9.217	3.214	0.00315
2	3.71	8.684	3.210	0.00304
3	3.84	8.348	3.194	0.00298
4	3.86	8.312	3.197	0.00294
5	3.40	9.464	3.206	0.00327
Average	/	/	3.204	0.00308

Note:

1.Test specimens were cut from the sample.

2.The test was carried out by SGS-CSTC Standards Technical Service (Shanghai) Co., Ltd.

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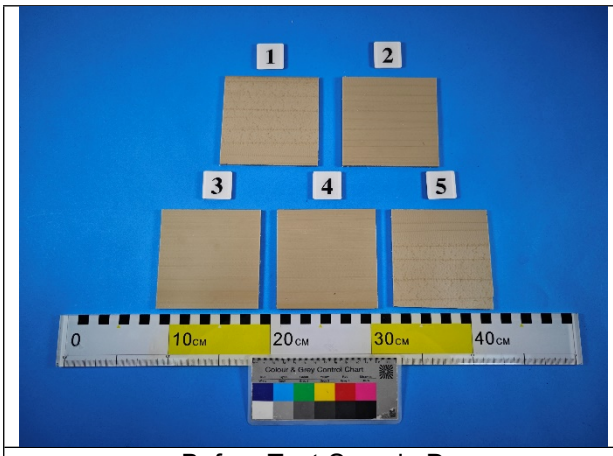
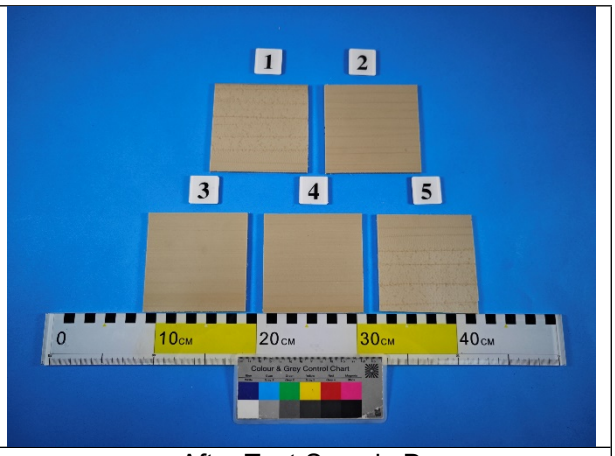
Test Photo:

	
Before Test-Sample D	After Test-Sample D

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
LCR Tester	E4980AL	P-576	2023-10-16	2024-10-15
Dielectric Test Fixture	16451B	P-576-1	2023-10-17	2024-10-16
Digital Calipers	0-150mm-0.01mm	P-698	2024-03-01	2025-02-28

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TEST REPORT

13.Test Item: Breakdown Voltage and Dielectric Strength

Sample Description: Sheet

Test Method: IEC 60243-1:2013

Test Condition:

Precondition: (23±2) ℃, (50±5) %RH, 24 h

Surround medium: Transformer oil at (23±2) ℃

Test electrode: Unequal electrodes (upper electrode Φ25 mm, lower electrode Φ75 mm)

Voltage type: AC 50 Hz

Mode of increase of voltage: Rapid-rise, 2000 V/s

Test Result:

Specimen No.	Thickness (mm)	Breakdown Voltage (kV)	Dielectric Strength (kV/mm)
1	3.68	37.99	10.32
2	3.67	37.43	10.20
3	3.69	37.12	10.06
4	3.68	37.62	10.22
5	3.67	37.69	10.27
Median	--	37.62	10.22

Note: Test specimens were cut from the sample.

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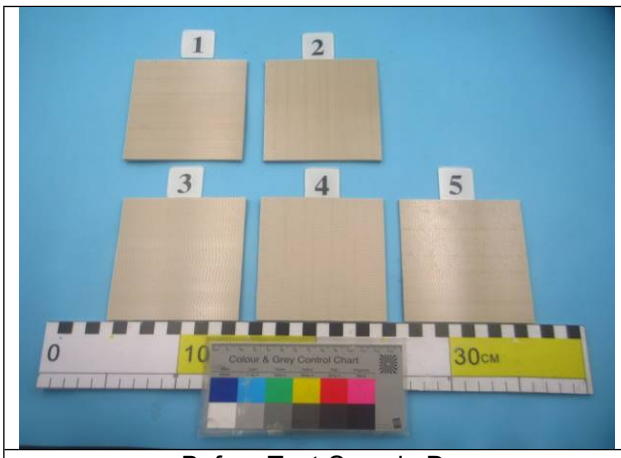	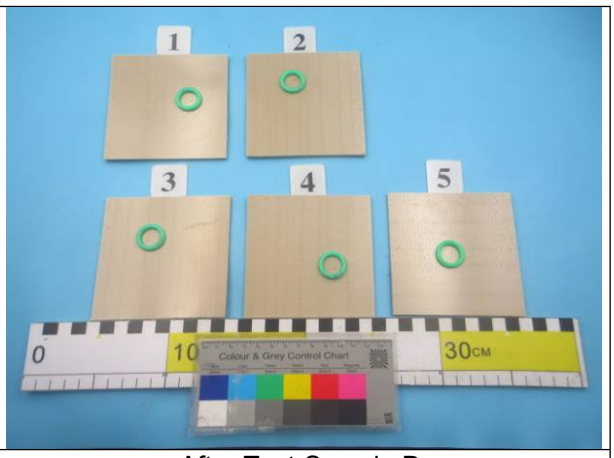
Before Test-Sample D	After Test-Sample D

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Dielectric Breakdown Voltage Tester	DJC-100KV	GZMR-PL-E154-01	2023-06-07	2024-06-06
Thickness meter	S112MXB	GZMR-RL-E115-01	2023-10-31	2024-10-30

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TEST REPORT

14.Test Item: Charpy Impact Strength

Sample Description: Sheet

Test Method: ISO 179-1:2023

Test Condition:

Specimen: ISO 179-1/1eA (Notch preparation: machining)

Specimen thickness: 3.72mm

The capacity of the pendulum: 1J

Impact speed: 2.9m/s

Span: 62mm

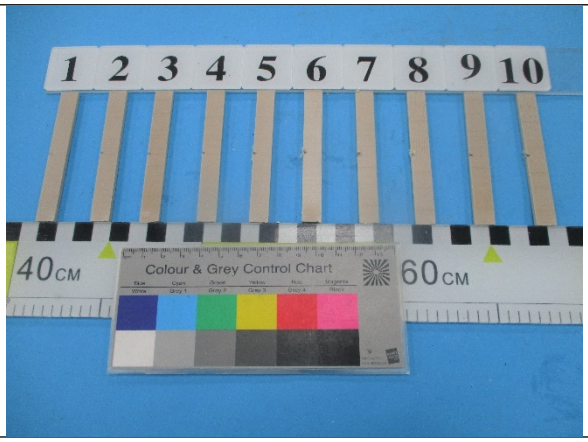
Lab Environmental Condition: (23±2) ℃，(50±5) %RH

Test Result:

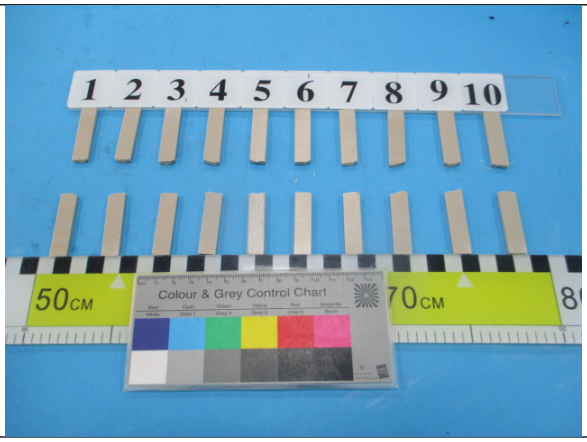
Test Item	Test Result
Charpy Notched Impact Strength	6.88kJ/m^2 C (complete break)

Note: Test specimens were cut from the sample.

Test Photo:



Before Test-Sample D



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TEST REPORT

Equipment Information:

Equipment	Model	Equipment No.	Calibration date	Next Calibration date
Impact Tester	HIT50P	GZMR-PL-E283	2023-06-26	2024-06-25

******** End of report********

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